

Ashraf Khan

Full-Stack Software Engineer · Quantum Software Engineer · HPC & Hybrid Systems Engineer

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🌐 **GitHub:** github.com/aashiq-parinda | 💼 **LinkedIn:** linkedin.com/in/ashrafksalim | 📱 **10+ Published Apps** (App Store & Play Store)

🎯 Professional Executive Summary

Accomplished **Full-Stack Software Engineer (5+ years experience, 10+ published commercial apps)** transitioning into **Quantum Software Engineering & Hybrid Classical-Quantum Systems (HPC × Qiskit)**. Combines production software engineering discipline (Python, Flutter, React Native), scalable cloud infrastructure, and enterprise security with original research in Transformer parameter/topology warm-starting for VQE and Quantum Architecture Search (QAS). Author of two published research preprints (Zenodo) and active upstream open-source code contributor to **Qiskit (IBM Quantum)**, **Mitiq (Unitary Fund)**, and **PennyLane (Xanadu)**.

🌐 Live Portals & Commercial Mobile Applications

🏢 Enterprise Web Portals & Ecosystems

- **Monitor Health AI (Primary Landing):** monitorhealth.ai — Commercial AI healthcare landing page & sales engine.
- **Remote Survey Portal:** survey.monitorhealth.ai — HIPAA-compliant remote health survey platform.
- **Institutional School Platform:** main.di10toj4buhn9.amplifyapp.com — Multi-tenant health tracking platform for educational institutions.

📱 Published Commercial Mobile Applications (App Store & Play Store)

- **MAXX AI Voice Assistant:** [App Store Link](#) — Voice AI assistant featuring custom wake-word & multi-advisor flows via Millis.ai.
- **Lead Yatra (Reliance / IndusInd):** [App Store Link](#) | [Google Play Link](#) — Enterprise field sales lead capture & banking venue booking.
- **Kokomo 24/7:** [App Store Link](#) — Scalable enterprise 24/7 operational service management platform.
- **FitWell | All-in-One:** [App Store Link](#) — HIPAA-compliant health platform with biometric auth, CDC/CMS reporting, and private Jitsi video consultations.

- **InstaQuote 2.0 by HDFC Life:** [App Store Link](#) | [Google Play Link](#) — Instant insurance quoting & premium calculation engine.
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Original Quantum Research & Zenodo Published Preprints

- **Ashraf Khan (2026).** *Transformer-Accelerated Variational Quantum Eigensolvers via Joint Architecture and Parameter Warm-Starting: A Multi-Molecule Empirical Study.* Zenodo Research Preprint. DOI: [10.5281/zenodo.22013110](https://doi.org/10.5281/zenodo.22013110).
 - Designed a 15,436-parameter pure-NumPy dual-head Transformer encoder predicting sparse entangling qubit-pair masks and warm-start rotation angles from molecular Hamiltonian Pauli tokens.
 - Achieved **66.7% to 83.1% 2-qubit (CX) gate count reduction** across H_2 , LiH , BeH_2 , and H_4 chain while maintaining energy convergence.
 - Formulated pre-registered falsifiable diagnostic criteria and direct gradient variance scaling analysis ($Var[\partial E / \partial \theta]$) identifying barren plateau onset at 8 qubits.
 - **Ashraf Khan (2026).** *Zero-Noise Extrapolation for Quantum Error Mitigation: A Systematic Reproduction & Discrepancy Study of Temme et al. (2017).* Zenodo Research Report. DOI: [10.5281/zenodo.21979332](https://doi.org/10.5281/zenodo.21979332).
 - Built complete mathematical reproduction of Temme et al. (PRL 119, 180509) implementing unitary gate folding $U \rightarrow U(U^\dagger U)^n$ and analytical Lagrange Richardson extrapolation.
 - Replicated Figures 1–3 and published quantitative discrepancy analysis under depolarizing and coherent amplitude damping noise models.
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Upstream Open-Source Code Contributions

- **IBM Quantum — Qiskit SDK (`qiskit.circuit.library`):**
 - **Upstream PR #16394 & Issue #16168:** Identified and resolved a root-cause bug in `MultiplierGate` high-level synthesis where custom register width configurations were incorrectly truncated during circuit decomposition. Merged upstream into Qiskit core.
 - **Unitary Fund — Mitiq (`mitiq.zne`):**
 - **Upstream PR #1866:** Improved naming conventions and API consistency in `noise_scaling_converter` module for zero-noise extrapolation workflows. Merged upstream.
 - **Xanadu — PennyLane (`pennylane.ops` & `pennylane.gradients`):**
 - **Upstream PRs #5818, #2667, #2649, #2062:** Authored QASM measurement parser integration, `PSWAP` gate operator, `IsingXY` interaction generator, and `Adjoint` methods for identity/barrier ops. Merged upstream.
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Professional Experience

Senior Mobile Application Engineer | Logicloop (Mumbai, India | Jan 2026 – Present)

- **Lead Yatra (Reliance IndusInd):** Architected cross-platform Flutter application optimizing field sales lead capture and venue booking at enterprise scale, securing banking data for IndusInd Bank operations.
- **MAXX AI Voice Assistant:** Developed end-to-end AI voice assistant featuring custom wake-word activation and multi-advisor conversational flows powered by Millis.ai.
- **Team Leadership & DevOps:** Led 3 developers, established GitHub Actions CI/CD pipelines (40% deployment overhead reduction), and enforced Level 2 VAPT security standards across financial flows.

Full-Stack Developer | Fitwell Technologies Inc. (Remote – VA, USA | Jun 2025 – Present)

- **Monitor Health AI Ecosystem:** Built commercial landing page (monitorhealth.ai), remote survey portal (survey.monitorhealth.ai), and multi-tenant institutional school platform (amplifyapp.com).
- **Private Cloud & Telehealth Video:** Deployed Flask microservices on AWS (EC2, S3, ELB), private Supabase cloud backend on EC2 for PHI data isolation, hosted private Jitsi Meet server for HIPAA-compliant consultations, and integrated Amazon Connect workflows.
- **Data Custodianship & Compliance:** Owned PHI data governance as data custodian, maintaining zero data breach posture and Level 2 VAPT security.

Lead Software Engineer | QuantGen Private Limited (India | Nov 2024 – Jun 2025)

- **Fitwell AI - All-in-One:** Led full-stack React Native development for HIPAA-compliant wellness platform with biometric auth, goal tracking, and CDC/CMS health reporting.
- **Kokomo 24/7:** Architected enterprise mobile solution for 24/7 service operations; built admin panel on AWS App Runner with AWS ECS orchestration and private Jitsi video servers.

Software Engineer | Logicloop (Mumbai, India | May 2024 – Oct 2024)

- **HDFC Life InstaQuote:** Built Flutter frontend for instant insurance quoting, financial premium calculations, and lead generation workflows.

Software Engineer | SoftProdigy (Remote – USA | Nov 2023 – Apr 2024)

- Developed loan lending application workflows in Flutter and implemented gRPC Client API libraries.

Senior Software Development Engineer | AKcess Labs Pvt. Ltd. (Remote – London, UK | Jan 2023 – Nov 2023)

- Built Flutter cross-platform applications with Flask backends; created the *CityShop* hyperlocal commerce application.

Software Developer | We3.Tech (Maharashtra, India | Aug 2021 – Jan 2023)

- **Mangal Keshav eKYC App**: Integrated eKYC and Digio eSign SDKs for digital onboarding.
- **Motilal Oswal MO Investor**: Delivered Add Funds module with banking API integrations and cryptographic security compliant with SEBI and RBI guidelines.

Unity Game Developer (Intern) | IDZ Digital Pvt. Ltd. (Mumbai, India)

- Prototyped 12+ interactive mobile games for KidloLand in Unity 3D.

Open-Source Portfolio Repositories & Notebook Suites

- **quantum-genai-warmstart** : Pure NumPy 15K-param dual-head Transformer predicting sparse entangling topologies (N_{CX} reduction) and VQE rotation angles θ_0 . Multi-molecule 10-seed benchmarks ($p < 0.05$).
- **quantum-hybrid-orchestration** : Classical-Quantum hybrid orchestration pipeline with smart backend router (Sim vs. QPU), achieving **99.78% execution cost reduction** across VQE and QAOA workloads.
- **quantum-circuit-search** : Quantum Architecture Search (QAS) sandbox implementing Genetic Algorithms for automated quantum circuit design (Bell, GHZ state preparation) and solving graph Max-Cut combinatorial optimization (expectations of 2.0 on 3-node graphs) vs. random brute-force.
- **quantum-applied-applications** : Applied Drug Discovery VQE Helix suite (`drug_discovery_vqe_helix.ipynb` — biomolecule target screening, 2D/3D conformers, Jordan-Wigner transform) and Quantum Hardware Benchmarking Suite (`quantum-validation-suite.ipynb` — Schrödinger dynamics, RB, T_1/T_2 noise, ZNE).
- **quantum-hpc-acceleration** : Tensor-sliced $O(2^N)$ statevector simulation engine using vectorized NumPy `einsum` contraction (scaling to 30 qubits).
- **quantum-error-correction** : Distance- d Surface Code stabilizer tableau extraction, Pauli syndrome detection, 9-qubit Shor code, 7-qubit Steane CSS code, and MWPM decoding via PyMatching.

Technical Skills & Core Competencies

- **Quantum Software & SDKs**: Qiskit, Qiskit Nature, PennyLane, Mitq, Stim, PyMatching, PySCF, Cirq.
- **HPC & Hybrid Quantum Middleware**: Vectorized NumPy (`einsum`), Tensor Slicing, Smart Backend Router (Sim vs. QPU), Hybrid Orchestration Pipelines, Parallel Dispatch.

- **Software & Systems Engineering:** Python / Flask, Flutter (Lead), React Native, Android (Java/Kotlin), iOS (Swift), REST, gRPC, Protobuf, WebSockets.
 - **Databases & Cloud Architecture:** Private Supabase Cloud on AWS EC2, PostgreSQL, Amazon Aurora, Docker, AWS (EC2, S3, ECS, App Runner, Amplify), GitHub Actions CI/CD.
 - **AI/ML & Quantum Optimization:** PyTorch, TensorFlow, Transformers, Quantum Architecture Search (QAS), Parameter-Shift Gradients, QNN, QSVC.
 - **Compliance & Security:** HIPAA, PCI-DSS, SEBI / RBI Guidelines, CDC/CMS Health Reporting, Data Custodianship & Governance, Level 2 VAPT, Biometric Auth, eKYC, Digio eSign.
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Education & Certifications

Education

- **B.Sc. in Computer Science** | Rizvi College, Mumbai University (2017 – 2020) — **CGPA:** 9.33 / 10.00
- **Vocational Computer Science / HSC (12th)** | M.H. Saboo Siddik Polytechnic, Mumbai (2015 – 2017) — **Percentage:** 75.80%

Certifications

- **IBM Quantum Credentials (Credly):**
- **Quantum Chemistry with Variational Quantum Eigensolver (VQE)** | Issued Oct 2025 | [Verify Badge](#)
- **Basics of Quantum Information** | [Verify Badge](#)
- **Quantum Machine Learning** | [Verify Badge](#)
- **Practical Introduction to Quantum-Safe Cryptography** | [Verify Badge](#)
- **Quantum Business Foundations** | [Verify Badge](#)
- **Credly Profile:** credly.com/users/ashraf-khan.729bf623
- **AI & Emerging Tech:** AI & Quantum Computing Mastery, Machine Learning with Python, Complete Quantum Computing Course.
- **Compliance:** Bioinformatics: Genomics & Proteomics; Healthcare API Compliance & HIPAA Data Practices; FinTech Regulatory Standards & PCI-DSS Payment Systems.